

Implicit, Related Rates & More: Guide

The big idea. Both topics on this page use one move: **differentiate an equation, using the chain rule on every variable that is itself changing.** In implicit differentiation, y depends on x . In related rates, everything depends on time t .

IMPLICIT DIFFERENTIATION

- 1 Differentiate both sides** with respect to x . Do not solve for y first.

2 Every y -term gets a $\frac{dy}{dx}$. $\frac{d}{dx}[y^n] = ny^{n-1} \frac{dy}{dx}$. Terms with only x differentiate as usual.

3 Products like xy need the product rule: $\frac{d}{dx}[xy] = y + x \frac{dy}{dx}$.
- 4 Gather** all the $\frac{dy}{dx}$ terms on one side, everything else on the other.

5 Factor out $\frac{dy}{dx}$ and divide to solve for it.

6 Plug in both x and y of the point to get the slope there.

Most common mistakes: (1) Forgetting the $\frac{dy}{dx}$ after differentiating a y -term. (2) Differentiating xy as just y (no product rule). (3) Plugging in only x : the point has a y -value too, so use both.

CHAIN RULE ON Y	PRODUCTS WITH Y	TANGENT LINES & SPECIAL SLOPES
y^n $ny^{n-1} \frac{dy}{dx}$	xy $y + x \frac{dy}{dx}$	Tangent line: $y - y_0 = m(x - x_0)$ with $m = \frac{dy}{dx}$ at the point.
e^y $e^y \frac{dy}{dx}$	$x^2 y^3$ $2xy^3 + 3x^2 y^2 \frac{dy}{dx}$	Horizontal tangent: numerator of $\frac{dy}{dx}$ is 0.
$\sin y$ $\cos y \cdot \frac{dy}{dx}$	Product rule first, then the chain rule on each y .	Vertical tangent: denominator of $\frac{dy}{dx}$ is 0.
$\ln y$ $\frac{dy/dx}{y}$		

RELATED RATES

- 1 Draw and label.** Give each changing quantity a variable. Label constants with numbers.

2 Given and find. Write rates as derivatives with units, like $\frac{dV}{dt} = 100 \text{ cm}^3/\text{s}$. Shrinking is negative.

3 Relate. One equation linking the variables. Eliminate extras first (similar triangles).
- 4 Differentiate with respect to t .** Every variable gets the chain rule: $\frac{d}{dt}[r^3] = 3r^2 \frac{dr}{dt}$.

5 Substitute the snapshot. Only now plug in the values at the given instant.

6 Solve and interpret. Answer in a sentence with units. The sign tells increasing (+) or decreasing (-).

Most common mistake: Plugging in numbers **before** Step 4. If you plug in $r = 5$ first, r becomes a constant and its derivative is 0. Differentiate first, then substitute.

AREAS & VOLUMES	TRIANGLES	READING THE PROBLEM
Circle $A = \pi r^2$	Right $x^2 + y^2 = z^2$	" How fast " or " rate " means a derivative.
Sphere $V = \frac{4}{3}\pi r^3$	Similar $\frac{b_1}{b_1} = \frac{b_2}{b_2}$	" At the instant when..." gives the snapshot values.
Cylinder $V = \pi r^2 h$	Cone tank $\frac{r}{h} = \frac{R}{H}$	Units show which rate: ft/s is $\frac{dx}{dt}$, ft ³ /min is $\frac{dV}{dt}$.
Cone $V = \frac{1}{3}\pi r^2 h$	Use similar triangles to remove a variable: a cone gives $r = \frac{R}{H}h$.	
Box $V = lwh$		

COMMON SETUPS AT A GLANCE

SITUATION	RELATE	DIFFERENTIATE IN T
Balloon or sphere	$V = \frac{4}{3}\pi r^3$	$\frac{dV}{dt} = 4\pi r^2 \frac{dr}{dt}$
Ladder against a wall	$x^2 + y^2 = L^2$	$x \frac{dx}{dt} + y \frac{dy}{dt} = 0$ (L is constant)
Two objects moving apart	$z^2 = x^2 + y^2$	$z \frac{dz}{dt} = x \frac{dx}{dt} + y \frac{dy}{dt}$
Cone tank (point down)	$V = \frac{1}{3}\pi r^2 h, r = \frac{R}{H}h$	Substitute r first, then $\frac{dV}{dt}$ in terms of $\frac{dh}{dt}$

L'Hôpital, Logs, Inverse Trig

Pick the method, then follow its steps.

L'HÔPITAL'S RULE (LIMITS)

- 1

Plug in first. A real number means you are done.
- 2

Check the form. Only $\frac{0}{0}$ or $\frac{\pm\infty}{\pm\infty}$ allows the rule.
- 3

Differentiate top and bottom separately (not the quotient rule): $\lim \frac{f}{g} = \lim \frac{f'}{g'}$
- 4

Take the limit again. If it is still $\frac{0}{0}$, apply the rule again.
- 5

Other forms: rewrite as a fraction first (table below).
- 6

Answer with a number, $\pm\infty$, or DNE.

FORM	WHAT TO DO
$\frac{0}{0}$ or $\frac{\infty}{\infty}$	Apply L'Hôpital directly.
$0 \cdot \infty$	Rewrite $f \cdot g$ as $\frac{f}{1/g}$ (or $\frac{g}{1/f}$), then apply the rule.
$\infty - \infty$	Combine into a single fraction over a common denominator.
$1^\infty, 0^0, \infty^0$	Let L be the limit, take $\ln$ of it, find that limit, then use $e^{(\cdot)}$.

Most common mistakes: (1) Using the quotient rule on top and bottom. (2) Using L'Hôpital when the form is **not** indeterminate. (3) Stopping at $\ln L$ without raising e to that power.

LOGARITHMIC DIFFERENTIATION

- 1

When: a variable in both the base and exponent ($x^x, x^{\sin x}$), or a long product or quotient.
- 2

Take $\ln$ of both sides and expand with log laws: $\ln(ab) = \ln a + \ln b, \ln \frac{a}{b} = \ln a - \ln b, \ln a^n = n \ln a$.
- 3

Differentiate implicitly. The left side becomes $\frac{dy}{dx}$.
- 4

Multiply by y to get $\frac{dy}{dx}$, and replace y with the original function.

INVERSE TRIG AND THE PRODUCT RULE

INVERSE TRIG (CHAIN RULE)	MORE INVERSE TRIG	PRODUCT RULE
$\arcsin u \quad \frac{u'}{\sqrt{1-u^2}}$	$\text{arccot } u \quad \frac{-u'}{1+u^2}$	Rule $(fg)' = f'g + fg'$
$\arccos u \quad \frac{-u'}{\sqrt{1-u^2}}$	$\text{arcsec } u \quad \frac{u'}{ u \sqrt{u^2-1}}$	Example $(x^3 \ln x)' = 3x^2 \ln x + x^2$
$\arctan u \quad \frac{u'}{1+u^2}$	$\text{arccsc } u \quad \frac{-u'}{ u \sqrt{u^2-1}}$	Two functions multiplied: first d-second plus second d-first.
Always multiply by u' , the derivative of the inside.	The "co" functions get a minus sign.	

WHICH METHOD?

THE PROBLEM HAS...	USE
x and y mixed together, y not isolated	Implicit differentiation
Two or more quantities changing in time, "how fast"	Related rates: relate, differentiate in t , then substitute
A limit that gives $\frac{0}{0}$ or $\frac{\infty}{\infty}$	L'Hôpital's Rule
A variable in the base <i>and</i> the exponent	Logarithmic differentiation
$\arcsin, \arctan, \dots$ of an expression	Inverse trig formula, then multiply by u'

Implicit & related rates in action

Draw it. Relate it. Differentiate. Then substitute.

Walkthrough 1: Slope and tangent line

IMPLICIT

 Find $\frac{dy}{dx}$ and the tangent line to $x^2 + y^2 = 25$ at $(3, 4)$.

1 DERIVE $2x + 2y \frac{dy}{dx} = 0$

2 SOLVE $\frac{dy}{dx} = -\frac{x}{y}$

3 SLOPE At $(3, 4)$: $\frac{dy}{dx} = -\frac{3}{4}$.

4 TANGENT $y - 4 = -\frac{3}{4}(x - 3)$, so $y = -\frac{3}{4}x + \frac{25}{4}$

Walkthrough 2: A product term

IMPLICIT

 Find $\frac{dy}{dx}$ for $x^2 + xy + y^2 = 7$ at $(2, 1)$.

1 DERIVE $2x + (y + x \frac{dy}{dx}) + 2y \frac{dy}{dx} = 0$. The xy term uses the product rule.

2 GATHER $x \frac{dy}{dx} + 2y \frac{dy}{dx} = -2x - y$

3 FACTOR $\frac{dy}{dx}(x + 2y) = -(2x + y)$, so $\frac{dy}{dx} = -\frac{2x + y}{x + 2y}$

4 SLOPE At $(2, 1)$: $-\frac{5}{4}$.

Walkthrough 3: Rates from a relation

IMPLICIT RELATED RATES

 $x^2y = 36$. When $x = 3$, $\frac{dx}{dt} = 1.5$. Find $\frac{dy}{dt}$.

1 DERIVE D/DT $2xy \frac{dx}{dt} + x^2 \frac{dy}{dt} = 0$ (product rule, both change).

2 FIND Y $9y = 36$, so $y = 4$.

3 PLUG IN $2(3)(4)(1.5) + 9 \frac{dy}{dt} = 0$, so $36 + 9 \frac{dy}{dt} = 0$.

4 SOLVE $\frac{dy}{dt} = -4$ (decreasing).

Walkthrough 5: Sliding ladder

RIGHT TRIANGLE

A 13 ft ladder slides away from a wall at 2 ft/s. How fast does the top fall when the base is 5 ft out?

1 RELATE $x^2 + y^2 = 169$

2 DERIVE $2x \frac{dx}{dt} + 2y \frac{dy}{dt} = 0$, so $x \frac{dx}{dt} + y \frac{dy}{dt} = 0$.

3 SNAPSHOT $x = 5$ gives $y = 12$: $5(2) + 12 \frac{dy}{dt} = 0$.

4 SOLVE $\frac{dy}{dt} = -\frac{5}{6} \approx -0.83$ ft/s: the top falls at about 0.83 ft/s.

Walkthrough 9: Rippling circle

AREA

 A ripple's radius grows at 3 ft/s. How fast is its area growing when $r = 10$ ft?

1 RELATE $A = \pi r^2$

2 DERIVE $\frac{dA}{dt} = 2\pi r \frac{dr}{dt}$

3 PLUG IN $2\pi(10)(3) = 60\pi \approx 188.5$ ft²/s

Walkthrough 4: Balloon

SPHERE

 A spherical balloon is inflated at 100 cm³/s. How fast is the radius growing when $r = 5$ cm?

1 RELATE $V = \frac{4}{3}\pi r^3$

2 DERIVE $\frac{dV}{dt} = 4\pi r^2 \frac{dr}{dt}$

3 PLUG IN $100 = 4\pi(25) \frac{dr}{dt}$

4 SOLVE $\frac{dr}{dt} = \frac{1}{\pi} \approx 0.318$ cm/s

Walkthrough 6: Conical tank

SIMILAR TRIANGLES

 A cone (point down) has radius 3 m and height 6 m. Water enters at 2 m³/min. How fast is the level rising when $h = 4$ m?

1 REMOVE R $\frac{r}{h} = \frac{3}{6}$, so $r = \frac{h}{2}$.

2 RELATE $V = \frac{1}{3}\pi \left(\frac{h}{2}\right)^2 h = \frac{\pi h^3}{12}$

3 DERIVE $\frac{dV}{dt} = \frac{\pi h^2}{4} \frac{dh}{dt}$

4 PLUG IN $2 = \frac{\pi(16)}{4} \frac{dh}{dt} = 4\pi \frac{dh}{dt}$

5 SOLVE $\frac{dh}{dt} = \frac{1}{2\pi} \approx 0.159$ m/min

Walkthrough 7: Shadow

SIMILAR TRIANGLES

A 6 ft person walks away from a 15 ft lamp at 4 ft/s. How fast does the tip of the shadow move, and how fast does the shadow lengthen?

1 SIMILAR TRI.S $\frac{s}{6} = \frac{x+s}{15}$, so $9s = 6x$ and $s = \frac{2}{3}x$.

2 TIP Tip is $x + s = \frac{5}{3}x$ from the lamp. $\frac{d}{dt} = \frac{5}{3}(4) = \frac{20}{3} \approx 6.7$ ft/s

3 LENGTH $\frac{ds}{dt} = \frac{2}{3}(4) = \frac{8}{3}$ ft/s

Walkthrough 8: Two moving objects

DISTANCE

Car A drives east at 60 mph and car B south at 40 mph from the same corner. How fast is the distance between them growing after 2 hours?

1 POSITIONS After 2 h: $x = 120$, $y = 80$.

2 RELATE $z^2 = x^2 + y^2$, so $z = \sqrt{20800} \approx 144.2$.

3 DERIVE $z \frac{dz}{dt} = x \frac{dx}{dt} + y \frac{dy}{dt}$

4 PLUG IN $\frac{dz}{dt} = \frac{120(60) + 80(40)}{144.2} \approx 72.1$ mph

L'Hôpital, logs, inverse trig, products

Check the form first, then pick the rule.

Walkthrough 10: L'Hôpital, once

0/0

$$\lim_{x \rightarrow 0} \frac{e^{2x} - 1}{x}$$

1 PLUG IN $\frac{e^0 - 1}{0} = \frac{0}{0}$: the rule applies.

2 DERIVE Top: $2e^{2x}$. Bottom: 1.

3 LIMIT $\lim_{x \rightarrow 0} \frac{2e^{2x}}{1} = \boxed{2}$

Walkthrough 11: L'Hôpital, twice

STILL 0/0

$$\lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2}$$

1 FIRST PASS $\frac{0}{0}$, so $\frac{\sin x}{2x}$.

2 STILL 0/0 Plug in: $\frac{0}{0}$ again. Apply the rule again.

3 SECOND PASS $\frac{\cos x}{2} \rightarrow \frac{1}{2}$

Walkthrough 12: A product form

0 · ∞

$$\lim_{x \rightarrow 0^+} x \ln x$$

1 FORM $0 \cdot (-\infty)$. Not a fraction yet.

2 REWRITE $x \ln x = \frac{\ln x}{1/x}$, which is $\frac{-\infty}{\infty}$.

3 APPLY $\frac{1/x}{-1/x^2} = -x$

4 LIMIT $\lim_{x \rightarrow 0^+} (-x) = \boxed{0}$

Walkthrough 13: L'Hôpital, infinity over infinity

∞/∞

$$\lim_{x \rightarrow \infty} \frac{x^2}{e^x}$$

1 FORM $\frac{\infty}{\infty}$: the rule applies.

2 FIRST PASS $\frac{2x}{e^x}$, still $\frac{\infty}{\infty}$.

3 SECOND PASS $\frac{2}{e^x} \rightarrow \boxed{0}$. Exponentials beat powers.

Walkthrough 14: Exponent form

1 TO THE ∞

$$\lim_{x \rightarrow \infty} \left(1 + \frac{1}{x}\right)^x$$

1 FORM 1^∞ . Let L be the limit and take $\ln$.

2 REWRITE $\ln L = \lim x \ln \left(1 + \frac{1}{x}\right) = \lim \frac{\ln(1 + 1/x)}{1/x} \left(\frac{0}{0}\right)$.

3 APPLY $\frac{-1/x^2}{1 + 1/x} = \frac{1}{1 + 1/x} \rightarrow 1$

4 UNDO LN $\ln L = 1$, so $L = \boxed{e}$

Walkthrough 15: Variable in the exponent LOG DIFFERENTIATION

 Find $\frac{dy}{dx}$ for $y = x^x$.

1 TAKE LN $\ln y = x \ln x$

2 DERIVE $\frac{dy/dx}{y} = 1 \cdot \ln x + x \cdot \frac{1}{x} = \ln x + 1$

3 MULTIPLY $\frac{dy}{dx} = \boxed{x^x(\ln x + 1)}$

Walkthrough 16: Another log-diff

X TO A FUNCTION

 Find $\frac{dy}{dx}$ for $y = x^{\sin x}$.

1 TAKE LN $\ln y = \sin x \ln x$

2 DERIVE $\frac{dy/dx}{y} = \cos x \ln x + \frac{\sin x}{x}$

3 MULTIPLY $\frac{dy}{dx} = x^{\sin x} \left(\cos x \ln x + \frac{\sin x}{x} \right)$

Walkthrough 17: Product / quotient

LOG DIFFERENTIATION

 Find $\frac{dy}{dx}$ for $y = \frac{x^3 e^{2x}}{(x+1)^2}$.

1 EXPAND $\ln y = 3 \ln x + 2x - 2 \ln(x+1)$

2 DERIVE $\frac{dy/dx}{y} = \frac{3}{x} + 2 - \frac{2}{x+1}$

3 MULTIPLY $\frac{dy}{dx} = \frac{x^3 e^{2x}}{(x+1)^2} \left(\frac{3}{x} + 2 - \frac{2}{x+1} \right)$

Walkthrough 18: Inverse trig chain rule

MULTIPLY BY U'

 Differentiate (a) $\arctan(3x)$ and (b) $\arcsin(x^2)$.

1 (A) $u = 3x, u' = 3: \frac{3}{1 + (3x)^2} = \frac{3}{1 + 9x^2}$

2 (B) $u = x^2, u' = 2x: \frac{2x}{\sqrt{1 - (x^2)^2}} = \frac{2x}{\sqrt{1 - x^4}}$

3 WATCH Square the **whole** inside, and never skip u' .

Walkthrough 19: Product rule with inverse trig

PRODUCT

 Differentiate $f(x) = x \arcsin(2x)$.

1 NAME $u = x$ and $v = \arcsin(2x)$, so $u' = 1$.

2 INVERSE TRIG $v' = \frac{2}{\sqrt{1 - 4x^2}}$ (inside $2x$ gives the 2).

3 PRODUCT RULE $f' = u'v + uv' = \arcsin(2x) + \frac{2x}{\sqrt{1 - 4x^2}}$